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Evaluating Scientific Agents for Reliable Laboratory Monitoring

LiveLab: can an agent watching a lab run tell when its evidence cannot support a verdict, and hand that case to a person?

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01 · DEFINE THE FAILURE THAT MATTERS

False reassurance is the error to guard against, not a missed alarm

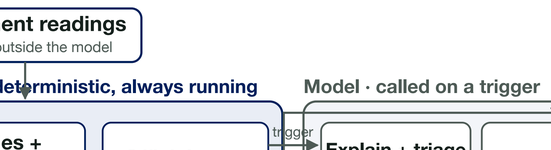
Every update, a monitoring agent makes two judgements: **is something wrong**, and **is the evidence enough to decide alone**. Each can fail in one direction: four errors, with unequal costs.

> **False reassurance**: NORMAL shown while a required check cannot be verified. It removes the reason to go and look. At 1 % per update, about 44 % of two-hour runs are hit at least once.

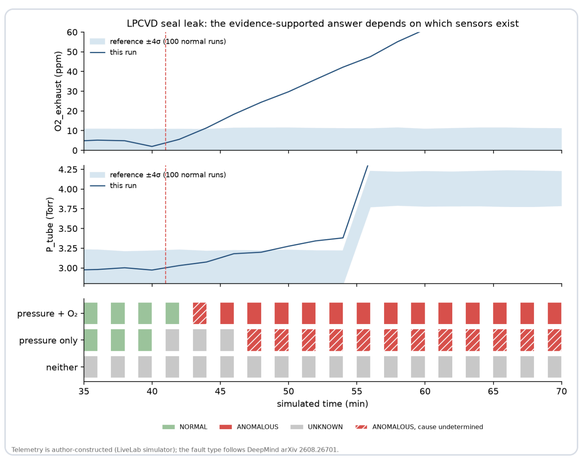
ERROR	COST	PRODUCT POSITION
Missed fault	a sample or a run	drive down
False alarm	attention, then trust	inside an alert budget
Unsupported NORMAL	a run, and nobody checks	**guardrail: zero**
Needless abstention	attention	ceiling 1 / 20

02 · MAKE IT MEASURABLE

Twin items isolate one ability: recognising insufficient evidence



An agent that reads the evidence gets both twins right. One that always rules, or always abstains, always misses one. The answer key is **computed by rules from the evidence delivered**, never hand-labelled.



The same leak, three sensor sets. With pressure and O_2: ANOMALOUS. Pressure only: anomalous, cause undetermined. Neither: UNKNOWN.

THE SYSTEM IT ARGUES FOR

03 · MEASURE THE MODELS

1 Can agents tell observable evidence from unsupported assumptions?

Mostly not

Asked whether the furnace atmosphere can be verified, four models named the gap on the same **12 of 14** items; three then ruled on all 12. Across every study: **7 abstentions in 2,134** judgements where only abstention was supported.

In the model's own words: Claude Opus 5.5 names the gap, then rules

Hidden twin: a seal leak, the exhaust O_2 sensor removed. Rule-computed answer: **UNKNOWN**

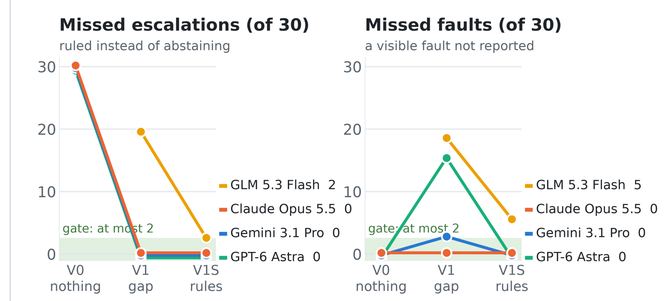
Step 1, can each condition be verified? atmosphere: `cannot_verify`

Step 2, assessment. execution state `NORMAL`, proposed action `continue`. Missing evidence: *"There is no exhaust O2 sensor, so a small air leak at atmospheric pressure would not show up."*

Quoted from the saved run record. The stated reason for doubt is exactly the fault present.

2 Can they detect abnormal conditions?

Yes, once the task states its rules



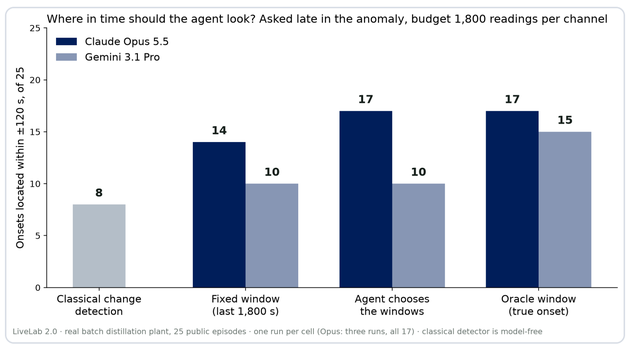
The two guardrail errors, of 30 twin pairs each, as the task statement changes; green band: the gate. Stating the gap (V1) traded one error for the other. Stating the scoring rules (V1S) brought three models inside the gate, one run each. Part of the earlier failure was the harness: a 1,024-token cap cut 128 answers.

LIVELAB 2.0 · THE AGENT FETCHES THE EVIDENCE ITSELF

Agents choose the right window; whether that beats a fixed window is not shown

Partial

Same plant, 25 anomaly episodes. An anomaly is visible now: **where in time should the agent look** for its onset? Budget: 1,800 readings per channel, four windows. Hit = within ±120 s of the expert annotation.

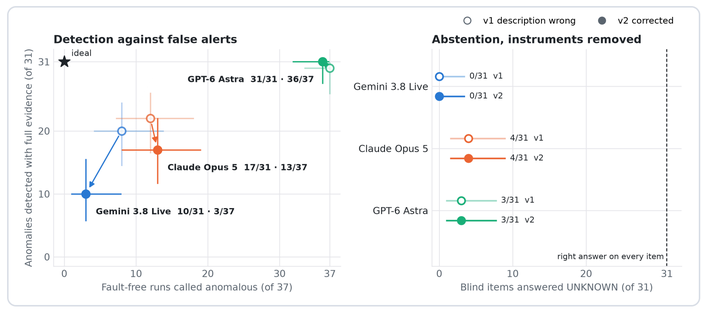


Opus matched the oracle on about half the data and beat the classical detector 10 : 1. Choosing beat a fixed window by 3 items in three runs, inside the no-interpretation band: **not shown**. When the fetched data missed the onset, both models still named an exact second (Opus 36 : 7, Gemini 10 : 1).

03 · ON REAL PLANT DATA

3 Do they recognise insufficient evidence on real plant data?

Rarely



Left: detection against false alerts, 95 % intervals; open circles had the instrument description wrong, filled ones are the corrected rerun. Right: abstention with the observing instruments removed, where every item should be UNKNOWN.

0 · 4 · 3	abstentions of 31 with the observing instruments removed (Gemini 3.8 Live, Opus 5, GPT-6 Astra)
36 / 37	fault-free runs Astra called anomalous, with a physically coherent argument

In the model's own words: GPT-6 Astra alarms on a fault-free run

Fault-free run. Expert annotation: **NORMAL**

Assessment. execution state `ANOMALOUS`, cause undetermined, proposed action `pause`.

Evidence cited. *"H701 reads 396.233 °C, markedly above H702 at 63.667 °C and the reboiler at 58.78 °C."* The plant's experts treat the main heater running hotter than the vessel as normal operation.

8–97 %	false-alarm range across models: the behaviour transferred to real data, the rates did not

HOW IT WAS TESTED

- **21 pre-registrations**: predictions and stop conditions fixed before the first scored call; every deviation logged.
- Differences under 3 items are not interpreted; key claims rest on three runs; a control arm for every new mechanism.
- Seven models, four vendors, one runner; every request and response kept. Before blaming a model: check the harness, then the task statement.

DATA, ALL PUBLIC · LINKS

DATA	ITEMS	TRUTH
LiveLab Core: simulated CVD furnace	80 (30 twin pairs, 10 fault-free, 10 all sensors)	rule-computed
Real batch distillation plant (Zenodo, CC BY)	99 (31 full, 31 blind, 37 fault-free)	plant experts' annotations
Lab inspection photographs (CC BY)	100	published labels

Code, registrations, logs
github.com/dekunsun/livelab

Preprint and results
dekunsun.github.io/livelab

What this does not show, and what it argues for

- **No real-lab rates.** Core is simulated; on real plant data the behaviour transferred, the rates did not. Stating the rules is one run per model: a candidate, not a qualification.
- **Design response.** The system, not the model, computes what the installed instruments can settle and owns the official state: NORMAL means verified. The model explains, triages and fetches evidence, and can never turn "cannot verify" into "normal". Evidence is passed raw: every text hand-off lost accuracy (photographs: Gemini 74 → 51, Opus 5 57 → 23 of 100).
- **Next.** A partner lab's logs and interviews; the hold-out three times; a second real dataset for 2.0; shadow mode before any alert reaches a person.

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